F16AG421-VS
VARIABLE SPEED DRIVE

RD WH BU
11,000V
FCF 100A
QAC 100A 100A 400A
BCT1 50 / 1A 2.5VA 5P20
BCT2 50 / 1A 2.5VA 5P20
BCT3 50 / 1A 2.5VA 5P20
BCN 50 / 1A 1.5VA 10P10
BAD
PFV1 RD RD RD
QCE 400A
EARTH BAR
X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13
2000-EL003-F16-10016 SHEET 14

SYSTEM INTERCONNECTION DIAGRAM 2000-EL005-B02-10001
L1 L2 L3
LINE SIDE INCOMING SUPPLY SHEET 6
OUTGOING MOTOR SUPPLY SHEET 6
T1 T2 T3
CABLE SCREEN EARTHED
M 3~
F16-AG-421
ROUGHER FLOTATION CELL 1 AGITATOR
3,300V
500 kW

F16-AG-421
MOTOR HEATER TERMINAL BOX
F16-HT-541
0.3kW
F16HT541-L
F16HT541-N
ANTI-CONDENSATION HEATER
MOTOR SPACE HEATER CIRCUIT SHEET 6
X81 (PART OF)
RTD TEMPERATURE MONITOR RELAY
SHEET 6

F16-AG-421
MOTOR RTD TERMINAL BOX
PT100 RTD INPUT CH1
PT100 RTD INPUT CH2
PT100 RTD INPUT CH3
PT100 RTD INPUT CH4
PT100 RTD INPUT CH5
PT100 RTD INPUT CH6
PT100 RTD INPUT CH7
PT100 RTD INPUT CH8
F16AG421A-RD
F16AG421A-WH
F16AG421A-BK
F16AG421B-RD
F16AG421B-WH
F16AG421B-BK
F16AG421C-RD
F16AG421C-WH
F16AG421C-BK
F16AG421D-RD
F16AG421D-WH
F16AG421D-BK
F16AG421E-RD
F16AG421E-WH
F16AG421E-BK
RTD
MOTOR WINDING RTD (RD PHASE) (TD1)
MOTOR WINDING RTD (WH PHASE) (TD3)
MOTOR WINDING RTD (BU PHASE) (TD5)
MOTOR BEARING RTD (DRIVE END) (BD10)
MOTOR BEARING RTD (NON DRIVE END) (BD20)

F16-SW-011 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10016 SHEET 7 3A & 3B
1139 1383
MV CONTACTOR ENABLE CIRCUIT SHEET 6
F16-SW-011 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10016 SHEET 10 4A
1179 1199
MV INPUT ENABLE CIRCUIT SHEET 6
F16-SW-011 11kV FEEDER VSD TRIP 2000-EL003-F16-10016 SHEET 10 3A
1219 1239
MV INPUT TRIP CIRCUIT SHEET 6
F16-SW-011 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10016 SHEET 8 5A & 5D
1259 1279
A30 (PART OF) TRIP
ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F16-DB-101
2000-F16-TLD-EL-10054
RD WH BU BK
R5 W5 B5
25A
415VAC
EARTH BAR
F16DB101-R5
F16DB101-W5
F16DB101-B5
F16DB101-BK5
VSD AUXILIARY POWER SUPPLY SHEET 6

F16-DB-201
2000-F16-TLD-EL-10057
RD BK
R1
10A
240VAC
EARTH BAR
F16DB201-R1
F16DB201-BK1
VSD UPS POWER SUPPLY SHEET 6

TO FLOTATION 11kV SWITCHROOM
F16-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F16-ETH-001
2000-F16-TER-IC-10002
CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6
X47 ETHERNET ADAPTOR RJ45
A2 (PART OF) DI-2B
24VDC CONTROL SHEET 6

SAFE TORQUE OFF CIRCUIT SHEET 6
F16AG421VS-1409
F16AG421VS-1419
F16AG421VS-1429
F16AG421VS-1439
F16AG421VS-1459
F16AG421VS-1469

F16AG421-LCS
LOCAL CONTROL STATION
LOS
START

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F16AG421-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.

LEGEND

- 11kV SWITCHBOARD TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL

PANEL A02

2000-F16-SCM-EL-10103_Rev1

F16-SW-011 F16-AG-421 rougher flot. cell 1 agitator VSD feeder - schematic diagram